

Tools for Observational Astronomy

Astronomy Activity

TEACHER NOTES

Level: Basic to Moderate level for Middle and High School.

Objective: The ultimate goal of the DSPIRA Astronomy Lessons Module is to give the teachers and students an opportunity to actually make observations in the sky using the DSPIRA Radio Horn Telescope. If the students are starting from ground zero in this discipline, then they will need to develop some basic tools, from where to look in their hometown skies to what is electromagnetic radiation and what can we learn from it.

Recommended Sequence of Activities

DSPIRA Activity	General Astronomy Students And Physics Students	Goal
	<u>The Nature of Light</u>	Astronomy is 99%+ about interpreting electro-magnetic radiation (EMR), sometimes simply called "light"
EM1	Intro to ElectroMagnetic Spectrum	An internet exploration to have the students engaged with creating an understanding of the Electro-Magnetic Spectrum along with a bit of history of its development.
EM1.*	The Importance of Spectroscopy	A good video from Crash Course Astronomy Series to introduce many aspects of light, with emphasis on the crucial importance of spectroscopy in the field of astronomy. The video link starts at discussion of Spectroscopy, but the whole video could be very helpful.
EM2	The Doppler Effect	The Doppler Effect is necessary for interpreting the spectrum that students will be gathering with the horn telescope.
	<u>Where to look?</u>	

DSPIRA Activity	General Astronomy Students And Physics Students	Goal
A2	Understanding Celestial Coordinate Systems	Basic activity to introduce a number of coordinate systems used in astronomy.
A3	Observing Coordinate Systems in Stellarium	An introduction to a good online planetarium program.
	<u>Intro to Radio Astronomy</u>	The “IRA” in “DSPIRA” is “In Radio Astronomy”. This video about Radio Astronomy created at MIT was considered to be one of the most informative videos relating to what DSPIRA and trying to get across to the students about Radio Astronomy.

Pre-Knowledge of Students:

- 1) Basic Algebra and calculator skills.
- 2) Graph interpretation

Key Terms:

ElectroMagnetic Radiation, Doppler Effect, Wavelength, Frequency, Spectroscopy, Radio Astronomy

Related Videos for Teacher:

[Crash Course Astronomy - Light](#) Works with EM 1 and EM 1.*

Teacher Preparation:

This grouping of activities doesn’t have a solid sequence to it, but it includes two major topics; the nature of light/ElectroMagnetic Radiation, and so general sky observational tools

These activities are building basic skills and each one can almost act in a stand-alone assignment fashion. Some of these activities show up in the other sequences

Activity EM 1 - Introduction to the ElectroMagnetic Spectrum

- This is a stand alone activity to introduce different aspects of EM radiation

Activity EM 1.* Importance of Spectroscopy

- Before the Doppler Shift can be fully understood, it is important to know what is being shifted. The Crash Course Astronomy - Light video, [starting at about 7:15](#), talks about how spectra are used to learn about the stars, including their relative motion.

Activity EM 2 - The Doppler Effect

- Doppler Effect lessons and activities are common, but this DSPIRA lesson actually uses data received by DSPIRA teachers using radio telescopes.
- If you (students) ask the internet “How fast is the Andromeda Galaxy heading towards us?”, a common answer is about 110 km/s. This refers to the Milky Way in general. Since the Solar System is moving quickly around the galaxy, an answer that includes the Earth's/Sun's motion has to be found.

Activity A2- Celestial Coordinate Systems

- Another foundation activity that works in a few DSPIRA Sequences ACT.A2 [Teacher Notes Page](#)

Activity A3- Observing Celestial Coordinates using Stellarium

- This is a more interactive experience for the students, building on the last activity on coordinate systems that was more informational.

Video - Radio Astronomy -

- If you are going to continue on with the DSPIRA Experience, then this is a great [Introduction video to Radio Astronomy](#) created by MIT students